Fintech Mobile App Onboarding Funnel Analysis

A end-to-end data analytics project. tracked users through an app onboarding funnel, find where they drop off, and recommend fixes backed by data.

The Business Problem

A fintech app ran a 6-month growth campaign and got 10,000 installs. Despite spending heavily on Paid Ads (35% of all users came from this channel), the overall conversion to first transaction was critically low. The questions I tried to answer:

- Where exactly in the funnel are users dropping off?
- Which acquisition channels are actually worth the spend?
- Does city tier (metro vs Tier 2/3) affect conversion?
- Are certain cohorts (signup weeks) performing better than others?
- What's driving the Day-7 retention problem?

Key Findings

Finding	Number
Overall conversion (install → first transaction)	**15.9%**
Biggest single drop-off step	**KYC: 35.8%**
Best acquisition channel	**Referral: 25.1%**
Worst acquisition channel	**Paid Ad: 7.0%**
Referral vs Paid Ad gap	**3.6× better**
Day-1 retention (vs 40–50% benchmark)	**9.6%**
Day-7 retention (vs 25–30% benchmark)	**5.4%**
Average D1 → D7 decay	**44.2%**
City Tier conversion difference	**None — all ~15.8%**

The two biggest problems: KYC friction is killing 1 in 3 users, and 90% of users don't come back the next day after installing.

Recommendations

1. **Fix KYC first** — it's the single biggest drop-off point (35.8%). Progressive disclosure UI (step 1 of 3 instead of one long form), Aadhaar OTP pre-fill, and real-time photo quality feedback would all help.
2. **Reallocate Paid Ad budget to Referral** — Paid Ad is the largest channel by volume (35% of users) but converts at only 7%. Referral converts at 25.1%. Even shifting 10% of Paid Ad budget to a referral incentive programme would meaningfully improve blended conversion.
3. **Days 2–6 re-engagement push sequence** — personalised notifications based on where the user stopped (e.g. "Your KYC is 80% done — complete it to unlock cashback") to recover the D1→D7 drop.
4. **Expand to Tier 2/3 markets without hesitation** — the data shows zero conversion penalty in smaller cities. Tier 3 actually has a slightly better Day-7 retention (5.8%) than Tier 1 (5.2%).

Project Structure

```
fintech-funnel-analysis/
|
 ├── data/
|   ├── users_clean.csv        # 10,000 users with attributes and funnel stage
|   ├── events_clean.csv       # 36,932 event-level records
|   ├── cohort_summary.csv     # 26 weekly cohorts with conversion + retention
|   ├── funnel_stages.csv      # 10 funnel stages with drop-off rates
|   ├── channel_summary.csv    # Channel-level performance summary
|   └── segment_summary.csv    # Tier, age, device, gender breakdown
|
 ├── notebooks/
|   ├── phase2_cleaning.ipynb       # Data validation and cleaning
```

```
|   ├── phase3_funnel_analysis.ipynb  # Funnel, segmentation, KYC deep-dive
|   ├── phase4_cohort_retention.ipynb # Weekly cohorts, D1/D7 retention
|   └── phase5_visualisations.ipynb  # Export-ready chart suite (14 charts)
|
├── charts/                  # All 14 exported PNG charts
|
├── dashboard/
|   └── fintech_funnel.pbix      # Power BI dashboard (3 pages)
|
├── fintech_simulator.py        # Data simulation script
└── README.md
```

Dataset

The dataset is fully synthetic but calibrated to published industry benchmarks:

- **AppsFlyer 2023** — 24% of installs never open the app (used for app_open drop-off)
- **Deloitte 2023** — 38% of users abandon during KYC (used for KYC drop-off calibration)
- **Mixpanel 2023** — Day-7 fintech retention is 25–30% for top apps (used as retention benchmark)

User attributes (age, city tier, gender, device) were generated using the Faker library with realistic Indian fintech user distributions.

Why synthetic? Real onboarding data from fintech companies is not publicly available — it's protected under India's DPDP Act and company NDAs. Synthetic data built on published benchmarks is industry-standard practice for portfolio projects, and is also how many companies do internal testing before launching features.

Tech Stack

Tool	Used For
Python (Pandas, NumPy)	Data simulation, cleaning, analysis
Plotly	Interactive and export-ready charts
Faker	Generating realistic Indian user profiles
Jupyter Notebook	Analysis workflow (inside VSCode)
Power BI Desktop	3-page interactive dashboard

Power BI Dashboard

Three pages:

Page 1 — Funnel Overview 10-stage funnel chart, step drop-off bar chart with conditional colouring (red = critical), 4 KPI cards, slicers for channel/device/city tier.

Page 2 — Cohort & Retention D1 vs D7 retention trend with industry benchmark bands, cohort heatmap (26 weeks), cohort size + conversion combo chart, cohort health scatter plot.

Page 3 — User Segments Channel conversion horizontal bar, volume vs quality scatter, city tier bars, age group bars, gender donut, KYC deep-dive panel.

How to Run

Python notebooks:

```
git clone https://github.com/YOUR_USERNAME/fintech-funnel-analysis

cd fintech-funnel-analysis

pip install pandas numpy plotly faker kaleido

# Open notebooks in order: phase2 → phase3 → phase4 → phase5
```

Power BI dashboard:

- Open dashboard/fintech_funnel.pbix in Power BI Desktop
- If data doesn't load, go to Home → Transform data → Data source settings → update the file paths to where you've saved the CSVs

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What I Learned

- Funnel analysis is more about asking the right questions than knowing the right tools — the KYC insight only emerged when I segmented by channel, not from the overall funnel view
- Synthetic data for portfolio projects is harder than it sounds — getting realistic drop-off patterns required reading actual industry reports, not just making up numbers
- Power BI's conditional formatting on bar charts is underrated for telling a data story quickly — the red KYC bar communicates the problem faster than any written insight
- Cohort analysis confirmed that the funnel problem is structural, not seasonal — no week solved it on its own, which rules out timing/campaign explanations